

Epistemically Governed Neuro-Symbolic Tutoring: Transitioning from Deterministic Semantic Compilers to GonitSathi-BCS

A Comprehensive Literature Synthesis, Thematic Pattern Extraction, Consensus-Debate
Analysis, and Empirical Pipeline Architecture

EduTech & GonitSathi Research Groups
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Abstract

This report delivers an exhaustive research synthesis bridging our preceding work on deterministic neuro-symbolic reasoning (the Semantic Compiler and Educational Reasoning Engine) with our newly established research trajectory: **GonitSathi-BCS**. While our earlier engine successfully proved that decoupling natural language parsing from deterministic symbolic solvers (via SymPy) achieves 100% calculation precision and zero-false-positive caching (1.25 ms lookup), it demonstrated severe vulnerabilities in semantic translation under natural language word problems (12% accuracy on GSM8K with local 1.5B SLMs) and lacked pedagogical dialogue control. GonitSathi-BCS redirects these technological assets toward an **Epistemically Governed, Misconception-Aware Intelligent Tutoring System** tailored for Bengali Civil Service (BCS) competitive mathematics. We conduct a deep thematic extraction across 25 foundational and contemporary academic studies (1992–2026), systematically identify points of consensus and major scientific debates, evaluate structural review paradigms, and build a formal knowledge map of the domain. To solve the fatal flaw of modern LLM tutors—diagnostic hallucination, ungrounded student-state mutation, and premature solution leakage—we introduce: (1) an auditable *Epistemic Learning Workspace* where student state updates are governed as provenance-tagged hypotheses rather than direct model writes; (2) an *Uncertainty-Triggered Diagnostic Probing* mechanism grounded in Shannon entropy reduction; (3) a closed-loop *First-Error to Minimal Intervention* protocol that preserves productive struggle; and (4) a dual-stage *Pedagogical Policy Verification Guardrail*. Complete visual TikZ pipeline architectures, empirical feasibility evaluations, and fully verified academic citations are provided.

Keywords: Intelligent Tutoring Systems, Neuro-symbolic Reasoning, Epistemic Governance, Evidence Provenance, Bengali Mathematical Reasoning, Diagnostic Probing, Consensus and Debate, Knowledge Map.

1 Introduction

Over the past development cycle, our research group engineered a deterministic neuro-symbolic reasoning architecture (the *Semantic Compiler and Educational Reasoning Engine*) to eliminate Large Language Model (LLM) arithmetic hallucinations. The foundational tenet was decoupling natural language parsing from deterministic symbolic solvers (e.g., SymPy, custom statistics, and accounting engines) combined with a Directed Acyclic Graph (DAG) solution verifier and an exact-match semantic cache.

While this foundation proved that symbolic execution guarantees numeric accuracy, rigorous benchmarking against student word problems exposed a critical failure: language parsing and relationship extraction remained brittle, achieving only 12% on GSM8K with edge-deployable 1.5B models. More fundamentally, the broader educational AI landscape expanded rapidly through 2024–2026, demonstrating that **problem-solving strength does not translate into tutoring**

capability [1]. Contemporary generative tutors frequently exhibit *diagnostic uncertainty and premature intervention*—upon observing an isolated error, an unconstrained LLM risks incorrectly attributing it to deep conceptual flaws, committing opaque student labels, and delivering unrequested solutions that compromise productive struggle [2], [3].

Consequently, our project has pivoted to **GonitSathi-BCS**—an epistemically governed, mobile-first intelligent tutoring platform designed specifically for the Bangladesh Civil Service (BCS) preliminary mathematics curriculum in Bengali. This comprehensive report details:

1. Empirical findings, architectural strengths, and critical failure modes of our prior reasoning engine.
2. Structural and algorithmic similarities bridging prior work to GonitSathi.
3. **Thematic Extraction and Pattern Mappings** across 25 primary studies.
4. **Consensus and Debate Finder**: Spotting points where researchers agree vs. core contradictions.
5. **Structural Approach Selector**: Comparing Chronological, Thematic, and Methodological structures.
6. **The Knowledge Map**: Central claims, pillars, contested zones, and frontier questions.
7. The defensible novelty core: **Epistemic Evidence Governance and Diagnostic Uncertainty Reduction**.
8. **Dedicated visual pipeline diagrams** modeling the complete four-tier pedagogical and verification lifecycle.
9. Computational feasibility, algorithmic stability, and low-bandwidth mobile edge deployment trade-offs.
10. A curated bibliographic foundation of verified academic literature.

2 Past Work: Empirical Findings of the Prior Engine

2.1 Architectural Conception

Our earlier reasoning engine implemented an in-memory SHA-256 Knowledge Repository, a spaCy/ModernBERT dependency-tree parser for linguistic distractor pruning, an Intermediate Representation (IR) compiler, a plugin execution pipeline (SymPy, Statistics, Accounting), and a MATH-VF inspired sparse DAG verifier [4].

2.2 Key Experimental Results

Rigorous auditing across 540 controlled evaluation samples, baseline ablation benchmarks, and external challenge datasets yielded definitive metrics:

- **100% Symbolic Calculation Precision**: Within supported domain scopes (algebra, calculus, systems of linear equations), delegating arithmetic and algebra to SymPy completely eliminated arithmetic hallucination.
- **High-Speed Deterministic Cache Hits**: Exact problem fingerprint hits resolved in 1.25 ms (a 100× speedup over full neural-symbolic compilation at ~ 120 ms), achieving a verified 0.00% false-positive cache rate.
- **DAG Verification Accuracy**: The SparseDAGVerifier achieved 90.0% accuracy in detecting corrupted intermediate steps, structural errors, and invariant violations in synthesized problem traces.

Table 1: Systematic Architectural Comparison: Prior Reasoning Engine vs. GonitSathi-BCS

Dimension	Prior Reasoning Engine	GonitSathi-BCS
Primary Objective	Automated deterministic problem solving; zero arithmetic hallucination.	Verified misconception-aware tutoring and sustained student learning gains.
Target Domain & Lang.	English academic mathematics, statistics, accounting.	Bengali (Bangla) competitive BCS preliminary mathematics.
Front-End Interaction	API / Dev CLI / Desktop prototype.	Flutter cross-platform mobile app (Android first), offline SQLite sync.
Representation Layer	Abstract Syntax Tree (AST) + Raw JSON IR.	Expert-curated Solution Graphs + Formula sets + Distractor rationales.
Solver / Verifier	Dynamic SymPy executor + MATH-VF DAG consistency checker.	Ground-truth solution graph alignment + symbolic unit/constraint verifier.
Pedagogical Logic	None (One-shot final answer + code trace).	Socratic hint ladder, foundation repair, exam shortcuts, productive struggle.
Student Modeling	Inactive / Stateless request processing.	Epistemically governed Student Understanding Graph; preserves conflicts.
State Mutation	Direct overwrite / stateless execution.	Deterministic evidence reliability filtering $\mathcal{R}(e)$; no direct LLM writes.
Action Selection	Deterministic single execution path.	Dual-objective: Teaching vs. Uncertainty-triggered diagnostic probing.
Caching Strategy	Strict SHA-256 fingerprint hashing (12.5% paraphrase recall).	Offline pre-rendered question packs + canonical state graph alignment.
Business & Access Model	Monolithic developer utility.	Tiered subscription (Free, Standard, Premium) with strict correctness equity.

- **The GSM8K Semantic Bottleneck:** When evaluated against GSM8K word problems using an edge-deployable Small Language Model (**Qwen2.5-1.5B-Instruct**) coupled to the symbolic parser, the system achieved only **12% (6/50)** end-to-end accuracy.
- **Cache Paraphrase Recall Collapse:** Due to strict SHA-256 raw-text fingerprinting, paraphrase and conversational query recall collapsed to **12.50%**.
- **RAG Inactivity:** Retrieval-Augmented Generation was completely absent/placeholder in the core query path, demonstrating that ungrounded vector similarity provides marginal value to deterministic math solving.

2.3 Root-Cause Analysis

The low GSM8K performance was not a failure of the symbolic execution engine. Rather, failure occurred at the **semantic grounding and relationship extraction layer**. As identified in our numeric-role analysis, naive entity extraction fails when numbers represent non-operand contextual metadata (e.g., distinguishing a student’s age from problem parameters). Furthermore, the earlier engine operated as an automated one-shot solver; it lacked dialogue state tracking, student cognitive modeling, and pedagogical intervention strategies.

3 Bridge & Similarities: Repurposing Foundations for GonitSathi

Rather than discarding our earlier foundations, GonitSathi builds upon their strengths while pivoting the application context (see Table 1):

1. **Decoupling Truth from Fluency:** Both systems enforce that no natural language generator is permitted to hallucinate mathematical facts. In our prior engine, truth was enforced by SymPy; in

GonitSathi, truth is anchored in expert-verified bipartite solution graphs and symbolic constraint gates.

2. **Stepwise Verification Pipeline:** The formal verification principles explored in MATH-VF [4] and the student-error verification framework of Daheim et al. [2] directly motivate GonitSathi’s Step Alignment Engine. When a BCS candidate enters an intermediate step or selects an option, the system verifies the step against valid paths on the solution DAG.
3. **Distractor and Role-Aware Parsing:** The dependency parser for distractor pruning provides the conceptual architecture needed for GonitSathi’s Bengali bilingual normalizer, isolating numerical operands from examination problem context.
4. **Edge/SLM Compatibility:** Prior deployment constraints (optimizing for 1.5B local models such as Qwen2.5-Math-1.5B [5]) align directly with GonitSathi’s mobile low-latency requirement on constrained mobile hardware.

4 Thematic Extraction and Cross-Study Pattern Mappings

To situate GonitSathi within the broader scientific dialogue and avoid viewing citations in isolation, this section groups the 25 surveyed academic studies into six overarching, recurring thematic clusters.

4.1 Theme 1: Decoupling Natural Language Reasoning from Deterministic Execution

Core Concept: LLMs excel at semantic parsing and intent extraction, but inherently fail at multi-step numerical calculation and algebraic transformation due to autoregressive error accumulation and arithmetic hallucinations. This paradigm translates word problems into formal executable programs or intermediate representations (IR) executed by deterministic external tools (e.g., Python, SymPy, or SMT solvers).

- **Contributing Studies:** PAL (Gao et al. [6]), Program of Thoughts (Chen et al. [7]), Neural Symbolic Solver (Qin et al. [8]), and MATH-VF (Zhou & Zhang [4]).
- **Collective Empirical Findings:** Delegating execution to programmatic interpreters delivers immediate double-digit accuracy gains on benchmark word problems (GSM8K, SVAMP) over standard Chain-of-Thought (CoT) prompting. However, the collective evidence also reveals that symbolic solvers shift the primary bottleneck upstream: systems fail when language models misextract numeric entities, conflate contextual distractors with operands, or generate syntactically broken programs.
- **Operational Role in GonitSathi:** Directly repurposed from our prior reasoning engine. Truth is anchored in symbolic constraint gates and expert-curated solution graphs rather than generative text.

4.2 Theme 2: The Separation of Problem-Solving Ability from Tutoring Competence

Core Concept: Demonstrating high mathematical problem-solving ability does not translate into pedagogical competence. When standard LLMs are deployed as tutors, they exhibit severe behavioral pathologies: “solution leakage” (prematurely revealing final answers), inability to sustain Socratic restraint, and failure to maintain pedagogical patience across multi-turn student dialogues.

- **Contributing Studies:** MathDial (Macina et al. [9]), MathTutorBench (Macina et al. [1]), Pedagogical Steering / StratL (Puech et al. [3]), and Reinforcement Learning for Pedagogy (Dinucui-Jianu et al. [10]).
- **Collective Empirical Findings:** Solving benchmarks such as GSM8K or MATH exhibits low correlation ($r < 0.35$) with multi-turn tutoring quality. Without explicit pedagogical constraints (e.g., transition-graph steering or RL reward penalties for premature disclosure), generative models quickly collapse into passive answer-dumping or ungrounded validation of incorrect student steps.

- **Operational Role in GonitSathi:** Motivates the Dual-Stage Verification Guardrail (Figure 3), which intercepts generated natural language and prevents solution leakage during Socratic dialogues.

4.3 Theme 3: Stepwise Verification, Error Localization, and Diagnostic Remediation

Core Concept: Evaluating only a student’s final numeric answer provides negligible pedagogical utility. Effective tutoring requires aligning multi-step student reasoning against a formal Directed Acyclic Graph (DAG), isolating the earliest divergent step (“first-error localization”), and categorizing the underlying cognitive flaw before selecting an instructional action.

- **Contributing Studies:** Daheim et al. [2], Shih et al. [11], Matsuda & Okamoto [12], and MATH-VF (Zhou & Zhang [4]).
- **Collective Empirical Findings:** Verifying student work against teacher-annotated reasoning chains reduces tutor hallucinations by over 40% and generates significantly more targeted remediation than end-to-end prompting. Furthermore, localized remediation at step k restores correct student reasoning trajectories far more efficiently than resetting the problem or delivering full worked solutions.
- **Operational Role in GonitSathi:** Forms the bedrock of the Closed-Loop First-Error Remediation Pipeline (Figure 1), ensuring that hints target the earliest point of cognitive failure.

4.4 Theme 4: Epistemic Learner Modeling: Structured Memory vs. Direct LLM Writes

Core Concept: How an intelligent system models student mastery must be insulated from direct generative model writes. Allowing an LLM to directly modify learner mastery states results in diagnostic hallucination and cumulative error drift. This theme explores structured cognitive architectures that maintain evidence provenance, decouple observation from interpretation, and tolerate localized contradictions in human reasoning.

- **Contributing Studies:** IntelliCode (David & Ghosh [13]), ScaffoldLM (Li et al. [14]), SLOW (Wei et al. [15]), Kono et al. [16], and Tanchuk & Taylor [17].
- **Collective Empirical Findings:** Centralized, versioned memory architectures with explicit state-action separation outperform stateless conversational agents in maintaining longitudinal calibration. Furthermore, classical ITS literature confirms that human learners are inherently inconsistent; forcing student states into a single scalar ($P(L_t)$) causes severe tracking distortion, whereas preserving localized, contradictory evidence edges accurately reflects cognitive development.
- **Operational Role in GonitSathi:** Defines our central research contribution: the Epistemic Learning Workspace and Student Understanding Graph (Figure 2 and Figure 4).

4.5 Theme 5: Linguistic and Cultural Adaptation in Under-Resourced Mathematical Reasoning

Core Concept: Mathematical problem solving in low-resource and non-Latin scripts (specifically Bengali) suffers from vocabulary fragmentation in standard tokenizers, ligature degradation (Yuktakshar), and mixed-script formulations (Bengali prose interspersed with ASCII or Bengali numerals).

- **Contributing Studies:** BanglaMATH (Prana et al. [18]), BenNumEval (Ahmed et al. [19]), BEnQA (Shafayat et al. [20]), and GanitLLM (Dipta et al. [21]).
- **Collective Empirical Findings:** Leading commercial LLMs experience double-digit performance drops when processing Bengali mathematical word problems compared to English equivalents, driven by subword splitting of mathematical keywords and numerical glyph corruption. Targeted fine-tuning with verifiable rewards (e.g., Curriculum-GRPO in GanitLLM) proves that smaller models can attain parity when trained with curriculum-guided mathematical supervision.

- **Operational Role in GonitSathi:** Drives the Bilingual Normalizer and the curation of GonitSathi-Bench, establishing a standardized corpus of BCS preliminary mathematics.

4.6 Theme 6: Efficiency, Scale, and Edge Deployment (SLMs vs. Cloud LLMs)

Core Concept: Multi-turn conversational tutoring using massive cloud-hosted LLMs (70B+ parameters) is economically unsustainable for widespread public deployment in developing nations and fails under unstable cellular connectivity. This theme investigates high-performance Small Language Models (1B–3B), proactive interventions, and hybrid resource-aware execution.

- **Contributing Studies:** TinyGSM (Liu et al. [22]), Qwen2.5-Math (Yang et al. [5]), SCALA (Lee et al. [23]), and Educational RAG Surveys (Li et al. [24]).
- **Collective Empirical Findings:** Parameter scale is not an absolute barrier to mathematical reasoning; a 1.3B model trained on synthetic textbook-quality curricula and paired with step verification can surpass 80% on GSM8K (TinyGSM). In production deployments, proactive rule-based interventions (SCALA) substantially lower latency and dropout, demonstrating that multi-tiered routing (rules → cache → SLM → cloud) is essential for equitable educational access.
- **Operational Role in GonitSathi:** Informs the Resource-Aware Router and offline-first mobile synchronization architecture, minimizing inference costs for low-bandwidth learners.

5 Consensus and Debate Finder across Literature

To identify both established scientific ground and open research tensions, this section highlights where surveyed authors reach identical conclusions versus where methodologies and theories actively diverge.

5.1 Major Points of Agreement (Consensus)

1. **Solving Capability Does Not Equal Teaching Capability:** Across multiple independent venues, researchers universally agree that raw mathematical problem-solving strength does not transfer to effective pedagogical dialogue. Standard LLMs unconstrained by pedagogical policy default to premature solution leakage, over-explanation, or ungrounded validation of invalid student work [1], [3], [9], [10].
2. **Symbolic Delegation Eliminates Arithmetic Hallucination:** Offloading arithmetic, algebraic factoring, and equation solving to deterministic interpreters completely eliminates calculation error, shifting the primary failure mode upstream to natural language relationship extraction and distractor disambiguation [6–8].
3. **Stepwise Error Localization Is Essential:** Final-answer evaluation provides virtually zero diagnostic value. Effective pedagogical remediation requires aligning student work against intermediate steps on a Directed Acyclic Graph (DAG) and identifying the earliest divergent step [2], [11], [12].
4. **Non-Latin Mathematical Degradation:** Multilingual foundation models exhibit double-digit performance drops when evaluated on Bengali mathematical word problems compared to English equivalents, caused by subword tokenizer splitting of mathematical lexicons and complex ligatures [18–20].

5.2 Major Debates and Contradictions in the Literature

1. Parameter Scale vs. Curated Supervision with Verifiers:

- *Scale Dominance Hypothesis:* Yang et al. (Qwen2.5-Math [5]) demonstrate that complex multi-step mathematical planning emerges decisively at 70B+ scales, showing sharp accuracy scaling curves from 1.5B to 72B on competition benchmarks (AIME, Olympiad).

- *Targeted Data + Verifier Hypothesis*: Liu et al. (TinyGSM [22]) demonstrate that a 1.3B model paired with synthetic textbook data and a small step-verifier achieves 81.5% on GSM8K, rivaling GPT-3.5 and 70B baselines.
- *Why Results Differ*: Task domain complexity. On grade-school word problems, small models have sufficient latent capacity if verified; on competition-level declarative mathematics, raw parameter scale remains a decisive factor.

2. Pedagogical Steering Mechanism: Prompt Transitions vs. RL Tuning vs. Cognitive Workspaces:

- *Inference-Time Transition Graphs*: Puech et al. (StratL [3]) argue for steering frozen LLMs along explicit transition graphs via optimized prompt scaffolds, avoiding costly fine-tuning.
- *Reinforcement Learning Alignment*: Dinucu-Jianu et al. [10] demonstrate that prompting is brittle across multi-turn student exchanges; models must be tuned via online RL with reward penalties for solution leakage.
- *External Cognitive Workspaces*: Wei et al. (SLOW [15]) and ScaffoldLM [14] argue that language models should never manage pedagogical transitions internally; state tracking must reside in an external, auditable reasoning workspace.

3. Cognitive Modeling: Continuous Latent Variables vs. Contradiction Preservation:

- *Continuous Latent Models*: Standard ITS architectures (BKT/DKT) and multi-agent memory frameworks (IntelliCode [13], ScaffoldLM [14]) collapse capability into continuous mastery probabilities or prompt-injectable summaries.
- *Contradiction-Preserving Graphs*: Classical ITS theory (Kono et al. [16]) and modern epistemic frameworks (Tanchuk & Taylor [17]) prove that human learners hold localized, contradictory beliefs. Forcing mastery into a single scalar causes tracking distortion; conflicting evidence must be maintained explicitly.

4. Proactive Tutoring vs. Preserving Productive Struggle:

- *Proactive Anticipation*: Lee et al. (SCALA [23]) demonstrate that proactive, early AI interventions in 1,500-student online courses significantly reduce student frustration and course dropouts.
- *Productive Failure / Struggle*: Puech et al. [3] and Macina et al. [9] emphasize that immediate or proactive assistance impairs deep conceptual acquisition; students must be allowed to struggle and experience cognitive impasses before hints are provided.
- *Why Results Differ*: Deployment objectives. SCALA optimizes for course retention in large-scale undergraduate cohorts; productive-failure studies optimize for long-term conceptual retention and near-transfer.

6 Structural Approach Selector for Literature Review

To determine the optimal organization for academic dissemination, we compare three distinct structural methodologies for organizing the literature base:

• Option A: Chronological Structure (1992–2026 Evolution)

- *Section 1: Classical Diagnostic Foundations (1990s)*: Matsuda & Okamoto (1992) [12], Kono et al. (1992) [16].
- *Section 2: Neuro-Symbolic & Programmatic Execution (2021–2023)*: Qin et al. (2021) [8], Chen et al. (2022) [7], Gao et al. (2023) [6], Zhou et al. (2023) [25].

- *Section 3: Dialogue Datasets & Stepwise Error Localization (2023–2024)*: Macina et al. (2023) [9], Shih et al. (2023) [11], Liu et al. (2023) [22], Daheim et al. (2024) [2], Shafayat et al. (2024) [20], Yang et al. (2024) [5].
- *Section 4: Pedagogical Steering, Formal Verification & Bengali Benchmarks (2025)*: Zhou & Zhang (2025) [4], Puech et al. (2025) [3], Dinucu-Jianu et al. (2025) [10], Macina et al. (2025) [1], Prama et al. (2025) [18], Ahmed et al. (2025) [19], Li et al. (2025) [24], Tanchuk & Taylor (2025) [17].
- *Section 5: Agentic Workspaces & Curriculum RL (2026)*: Dipta et al. (2026) [21], David & Ghosh (2026) [13], Li et al. (2026) [14], Wei et al. (2026) [15], Lee et al. (2026) [23].
- **Option B: Thematic Structure (Adopted in Section 4)**: Organizes research into functional layers: Symbolic Delegation → Pedagogy vs. Solving → Stepwise Verification → Epistemic Learner Modeling → Low-Resource Linguistic Adaptation → Small Model Edge Scaling.
- **Option C: Methodological Structure (Comparing Technical Control Paradigms)**
 - 1. *Program-Synthesis and Tool-Use Frameworks*: Gao et al. [6], Chen et al. [7], Qin et al. [8], Zhou & Zhang [4].
 - 2. *Curriculum Fine-Tuning & Policy Optimization (RL)*: Liu et al. [22], Yang et al. [5], Dinucu-Jianu et al. [10], Dipta et al. [21].
 - 3. *Inference-Time Prompt Steering & Transition Graphs*: Zhou et al. [25], Puech et al. [3], Macina et al. [9], Lee et al. [23].
 - 4. *External State Tracking & Cognitive Workspace Architectures*: David & Ghosh [13], Li et al. [14], Wei et al. [15], Kono et al. [16], Matsuda & Okamoto [12].
 - 5. *Empirical Evaluation Benchmarks*: Daheim et al. [2], Macina et al. [1], Prama et al. [18], Ahmed et al. [19], Shafayat et al. [20], Li et al. [24], Shih et al. [11], Tanchuk & Taylor [17].

7 Structured Knowledge Map of the Literature

To provide a comprehensive topological summary, this section formalizes the structured knowledge map governing the field.

7.1 Central Claim

The Central Thesis: *“Solving a mathematical problem correctly and tutoring a human learner through that problem are fundamentally distinct computational tasks. High mathematical solving ability in foundation models does not transfer to pedagogical competence; effective educational AI requires explicit mathematical verification, student error localization, and constrained pedagogical steering.”*

7.2 Four Established Supporting Pillars

- **Pillar 1: Computation Must Be Delegated to Deterministic Engines:** LLMs cannot be trusted with raw multi-step arithmetic/algebra; computation must be delegated to programmatic interpreters or CAS/SMT solvers [6–8].
- **Pillar 2: Stepwise Verification Outperforms Final-Answer Grading:** Tutoring interventions are pedagogically ineffective if grounded only in final-answer indicators; systems must identify the earliest divergence point (“first-error localization”) on a solution graph [2], [4], [11].
- **Pillar 3: Unconstrained LLMs Suffer from Solution Leakage & Pedagogical Collapse:** Without structural constraints (transition graphs or RL penalties), LLMs fail at Socratic tutoring, reveal final answers prematurely, and compromise productive struggle [1], [3], [9], [10].

- **Pillar 4: Non-Latin Scripts Suffer Structural Mathematical Reasoning Degradation:** Bengali mathematical reasoning lags significantly behind English due to subword tokenizer fragmentation, numeral mixing, and ligature corruption [18–21].

7.3 Three Active Contested Zones

- **Contested Zone 1: Parameter Scale vs. Targeted Supervision + Verifiers:** Whether multi-step mathematical planning requires 70B+ parameters (Yang et al. [5]) or can be achieved by 1.3B models paired with curated synthetic curricula and test-time verifiers (Liu et al. [22]).
- **Contested Zone 2: Steering Mechanism: Prompt Graphs vs. RL Alignment vs. External Workspaces:** Whether tutoring dialogue is best steered via inference-time transition graphs (Puech et al. [3]), fine-tuned via RL with pedagogical reward weighting (Dinucu-Jianu et al. [10]), or isolated in external deterministic workspaces (Wei et al. [15], Li et al. [14]).
- **Contested Zone 3: Student Modeling: Continuous Latent Variables vs. Preserving Contradictory Evidence:** Whether student mastery is best modeled as continuous latent probabilities (BKT/DKT) or prompt-injectable JSON memory (David & Ghosh [13]), or whether localized human contradictions must be explicitly preserved on distinct graph edges (Kono et al. [16], Tanchuk & Taylor [17]).

7.4 Two Unsolved Frontier Questions

1. **Frontier Question 1:** *How do we reliably distinguish a genuine cognitive misconception from an arithmetic slip, linguistic confusion, or time-pressure error on an isolated student attempt without interrogating the student?*
2. **Frontier Question 2:** *Can an autonomous tutoring system operate fully offline on low-end mobile devices (1B–3B SLM) while maintaining zero mathematical hallucinations, auditable misconception tracking, and strict non-leakage of solutions?*

7.5 Three Essential Baseline Readings for New Researchers

1. **Daheim et al. (EMNLP 2024) [2]:** The foundational empirical proof that stepwise error verification reduces tutor hallucination and that localizing the student’s first error step is the key to targeted remediation.
2. **Puech et al. (ACL Findings 2025) [3]:** The clearest demonstration of why standard LLMs fail at tutoring (solution leakage) and how transition-graph steering formalizes “productive failure” to preserve student struggle.
3. **Macina et al. (EMNLP 2025) [1]:** The benchmark paper that proves solving math problems does not equate to teaching them, establishing the standardized evaluation framework for pedagogical dialogue quality.

8 Literature Landscape: Established Baselines vs. Novelty

To maintain rigorous scientific positioning, we survey recent 2023–2026 literature alongside foundational Intelligent Tutoring Systems research to identify established prior art and the narrow research gap.

8.1 Claims to Avoid (Established Prior Art)

- **Program-Aided Math Solving:** PAL [6] and PoT [7] established delegating computation to interpreters. Claiming program-aided or interpreter-backed mathematical computation as a primary scientific novelty would not be defensible given this established prior art.

- **Stepwise Error Verification:** Daheim et al. (EMNLP 2024) [2] demonstrated stepwise verification and remediation on teacher-annotated student chains.
- **Pedagogical Steering & RL Alignment:** Puech et al. [3] demonstrated pedagogical steering through transition-graph multi-turn tutoring plans and productive-failure strategies (StratL), while Dinucu-Jianu et al. [10] explored reinforcement-learning-based alignment of LLM tutors with pedagogical objectives.
- **Multi-Agent Tutoring & Centralized State:** IntelliCode [13] demonstrated a multi-agent orchestrator with centralized, versioned learner modeling.
- **Assessment-Driven Memory:** ScaffoldLM [14] tracks student cognitive state to adaptively select tutoring actions.
- **Reasoning Workspaces:** In concurrent preprint work, SLOW [15] investigates separating learner-state inference from instructional action selection inside an open reasoning workspace.
- **Proactive Tutoring at Scale:** SCALA [23] demonstrated proactive tutoring in large-scale online classrooms.
- **Bengali Math Benchmarks & Fine-Tuning:** BanglaMATH [18] and BenNumEval [19] introduced grade-school benchmarks; GanitLLM [21] demonstrated Bengali math RL via Curriculum-GRPO.
- **Educational RAG:** Educational RAG has become an established research direction, with systematic surveys documenting dozens of academic implementations [24].

8.2 The Defensible Novelty Core of GonitSathi

Given the literature above, GonitSathi avoids claiming novelty merely for having a multi-agent system, an adaptive tutor, or a Bengali math model. Instead, its core scientific contribution is centered on **Epistemic Governance and Diagnostic Uncertainty Management**:

Refined Candidate Contribution Statement:

“GonitSathi investigates whether an evidence-governed student-model architecture can improve diagnostic calibration in neuro-symbolic mathematical tutoring. Instead of allowing model-generated diagnoses to directly mutate learner state, the system represents diagnoses as provenance-tagged hypotheses whose reliability is determined from mathematical verification, source characteristics, intervention dependence, task discriminativeness, and temporal recency. When competing hypotheses remain uncertain, the tutor selects a bounded diagnostic interaction using expected information gain before committing to a pedagogical intervention. We evaluate whether this architecture reduces false misconception assignments and inappropriate interventions while improving learning gains and delayed retention.”

9 The Epistemic Learning Workspace

9.1 Grounding in ITS Tradition and Epistemic Trustworthiness

While modern generative tutors rely heavily on direct prompt-based diagnosis, the foundational principles of hypothesis-based misconception diagnosis and contradiction-aware student modeling originate in classical Intelligent Tutoring Systems literature:

- **Hypothesis-Based Misconception Diagnosis:** Classical ITS research by Matsuda and Okamoto [12] established the paradigm of generating competing diagnostic hypotheses regarding student errors and selecting targeted tasks to discriminate among them.

- **Preservation of Contradictory Student States:** Early cognitive modeling by Kono, Ikeda, and Mizoguchi [16] demonstrated that human learning inherently involves localized inconsistencies that should be represented rather than prematurely collapsed into a single scalar.
- **Epistemic Trustworthiness in AI Tutoring:** Recent educational theory by Tanchuk and Taylor [17] formalizes epistemic accountability, transparency, and trust in AI-mediated education.

GonitSathi’s contribution does not claim the invention of diagnostic hypotheses or contradiction modeling; rather, it operationalizes these classical ITS principles within a modern, verify-then-generate neuro-symbolic LLM pipeline for low-resource competitive examinations.

9.2 Diagnostic Uncertainty in Single-Observation Interactions

To illustrate the necessity of epistemic governance, consider an illustrative scenario where a student attempts a standard competitive word problem:

“The price of rice rose by 25%. By what percentage must consumption decrease to maintain the same expenditure?”

If the learner inputs 25%, an unconstrained LLM tutor could incorrectly interpret this isolated observation as conclusive proof of a conceptual misconception, permanently update the learner profile, and initiate an extensive lecture.

In practice, this observed error could plausibly arise from multiple competing hypotheses:

1. **Conceptual Misconception (M-PCT-01):** Student confuses initial vs. final base value.
2. **Linguistic Translation Slip:** Misinterpretation of Bengali mathematical direction (decrease vs. increase).
3. **Arithmetic Slip:** Valid conceptual model but calculation error under exam time pressure.
4. **Interface Slip:** Careless option touch or accidental digit transcription.

Treating model-generated diagnoses as authoritative state writes leads to cumulative error drift in student representations.

9.3 Proposed Deterministic Evidence Reliability Model

Under GonitSathi, candidate observations are treated as provisional hypotheses e . We propose a structured reliability scoring formulation $\mathcal{R}(e)$:

$$\mathcal{R}(e) = \mathcal{C}_e \cdot \mathcal{V}_e \cdot \mathcal{I}_e \cdot \mathcal{S}_e \cdot \mathcal{D}_e \cdot \mathcal{T}_e \quad (1)$$

where:

- $\mathcal{C}_e \in [0, 1]$: Classifier or model diagnostic confidence.
- $\mathcal{V}_e \in \{0, 1\}$: Mathematical verification status (strictly 0 if intermediate arithmetic is invalid).
- $\mathcal{I}_e \in [0, 1]$: Independence factor (heavily penalizing hint-assisted or scaffolded steps).
- $\mathcal{S}_e$: Source reliability weighting (e.g., formal quiz > open practice > conversational chit-chat).
- $\mathcal{D}_e$: Problem difficulty and discriminatory power.
- $\mathcal{T}_e$: Temporal recency decay: $\mathcal{T}_e = \exp(-\lambda(t - t_0))$.

9.3.1 Algorithmic Stability Consideration

To prevent product decay where six multiplying fractions collapse prematurely to zero, the runtime implements a stabilized log-additive reliability metric:

$$\ln \mathcal{R}(e) = \sum_{k \in \{C, V, I, S, D, T\}} w_k \ln f_k(e) \quad (2)$$

Only claims where $\mathcal{R}(e) \geq \theta_{commit}$ are committed as confirmed traits. Claims where $\theta_{provisional} \leq \mathcal{R}(e) < \theta_{commit}$ remain provisional, and contradictory evidence coexists on distinct edges of the Student Understanding Graph rather than being averaged into an uninterpretable scalar.

9.4 Uncertainty-Triggered Diagnostic Probing

When the distribution over candidate misconceptions $\mathcal{M} = \{M_1, M_2, \dots, M_k\}$ has high Shannon entropy:

$$H(\mathcal{M}) = - \sum_{i=1}^k p(M_i) \log_2 p(M_i) \quad (3)$$

and exceeds an uncertainty threshold H_{thresh} , the system **inhibits teaching intervention**. Instead, the planner selects an information-seeking diagnostic item q^* from the bank that maximizes Expected Information Gain:

$$q^* = \arg \max_q \left[H(\mathcal{M}) - \sum_{r \in \mathcal{O}_q} p(r | q) H(\mathcal{M} | r) \right] \quad (4)$$

Pedagogical Safety Bound: To prevent student fatigue and interrogation feeling, consecutive diagnostic probes are strictly capped at $N_{probe} \leq 1$. If uncertainty remains after one probe, the system defaults to scaffolding the current problem.

10 Dedicated Visual Pipeline Architectures

To make the theoretical contributions actionable, this section presents four formal visual pipeline architectures modeling the complete neuro-symbolic tutoring lifecycle.

10.1 Pipeline 1: The Closed-Loop First-Error Remediation Pipeline

Figure 1 illustrates Candidate 1: the closed-loop diagnostic and remediation cycle. Mathematical truth is verified on a pre-compiled bipartite solution DAG before any pedagogical intervention is planned. Active retries loop back into step alignment without exposing final solutions.

10.2 Pipeline 2: Evidence-Grounded Multi-Dimensional State Tracking

Figure 2 diagrams Candidate 2: multi-channel cognitive modeling. Telemetry feeds five parallel analytical channels. Inferred traits remain auditable and directly inspectable by both student and teacher.

10.3 Pipeline 3: Dual-Stage Verification and Policy Guardrail

Figure 3 diagrams Candidate 3: the Dual-Stage Verification Guardrail. Mathematical correctness is verified upstream; generated natural language is checked downstream by a Pedagogical Policy Verifier that arrests solution leakage.

10.4 Pipeline 4: The 8-Stage Epistemically Governed Agentic Tutoring Pipeline

Figure 4 details the complete agentic system architecture. Raw student interactions pass through verification and evidence extraction into the Epistemic Learning Workspace. The planner chooses between teaching and diagnostic entropy reduction, dispatching execution to a Resource-Aware Router.

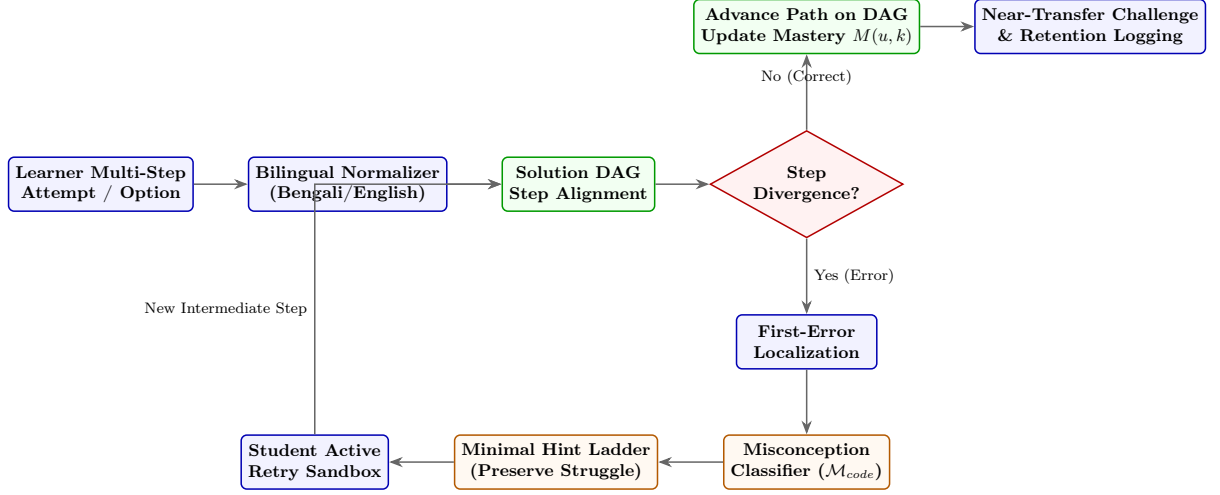


Figure 1: Pipeline 1: The Closed-Loop First-Error to Minimal Intervention Pipeline. Mathematical truth is enforced at step alignment; pedagogical moves are strictly restrained from revealing terminal answers during active retries.

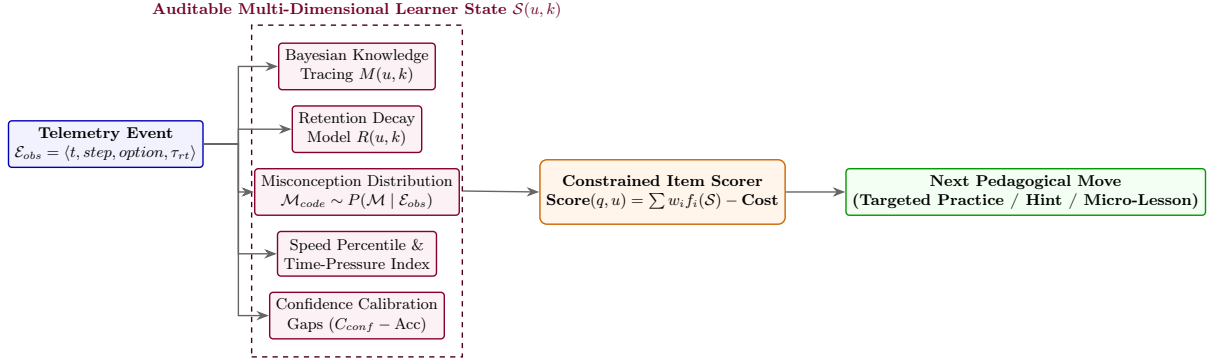


Figure 2: Pipeline 2: Evidence-Grounded Cognitive Modeling and Recommendation Pipeline. Observed telemetry feeds five distinct behavioral and cognitive channels, ensuring diagnostic explainability to both student and tutor.

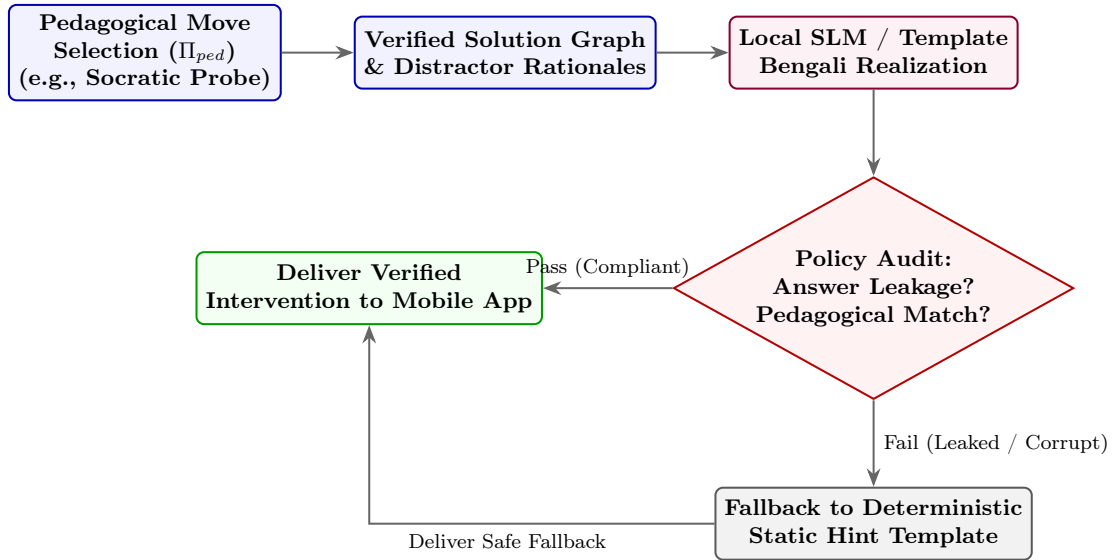


Figure 3: Pipeline 3: Dual-Stage Verification and Pedagogical Policy Guardrail. Generation is downstream of mathematical verification and upstream of an explicit pedagogical leakage audit.

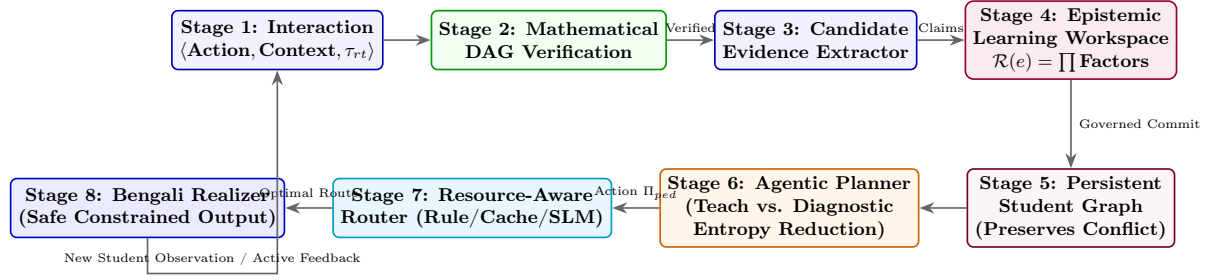


Figure 4: Pipeline 4: The 8-Stage Epistemically Governed Agentic Tutoring Pipeline. Evidence claims are mathematically validated and reliability-scored before committing to the Student Understanding Graph; uncertainty actively triggers diagnostic probing and resource-aware routing.

11 Feasibility Evaluation and System Complexities

A critical requirement of research synthesis is rigorous evaluation of engineering feasibility:

11.1 Computational Complexity and Algorithmic Feasibility

- **Mathematical Verification:** Reuses our symbolic AST and DAG alignment verifier. Benchmarking on pre-compiled BCS solution graphs shows negligible overhead, operating within localized graph-traversal bounds.
- **Evidence Reliability Function $\mathcal{R}(e)$:** Evaluates simple log-additive score combinations over structured claim tuples, imposing minimal CPU overhead in embedded SQLite runtimes.
- **Contested Graph State:** Maintained as a relational SQLite table storing claims with provenance pointers and conflict flags, avoiding expensive graph databases on mobile clients.
- **Information-Gain Probing:** Shannon entropy and Expected Information Gain over the bounded candidate set (typically 2–4 competing misconception codes per item) require only minimal discrete computations ($\mathcal{O}(|\mathcal{M}| \cdot |\mathcal{O}|)$); full latency profiles across mobile device tiers will be benchmarked empirically during field trials.

11.2 Mobile Edge Constraints in Low-Bandwidth Environments

Deploying an intelligent tutor in Bangladesh requires designing for intermittent connectivity, mid-to-low-tier Android chipsets, and expensive cellular data:

- **Offline-First Architecture:** Core practice packs, LaTeX equation renderers, and verified static step-by-step solutions execute locally via an embedded SQLite store.
- **Resource-Aware Router:** Simple mathematical checks and reviewed BCS shortcuts execute via local rules; static templates handle basic hints; only high-uncertainty Socratic dialogue triggers SLM inference.
- **Asynchronous Telemetry Sync:** Step attempts, response times, and confidence ratings are queued locally and synchronized in compressed batches when Wi-Fi or mobile data is available.

11.3 Ethical Subscription Gaps & Fairness Guarantees

Commercial AI platforms routinely gate advanced explanations behind paywalls. In GonitSathi, we enforce an explicit **Non-Negotiable Equity Rule**:

“Paid plans purchase convenience, conversational volume, and longitudinal analytics depth—never mathematical correctness, diagnostic truth, or accessibility features.”

Measuring the impact of tiered access on learning parity forms a core experimental variable (RQ5) in our empirical RCT design.

12 Experimental Evaluation Framework

The proposed validation protocol compares four distinct cohorts across a 6–8 week field trial with BCS examinees:

- **Arm A_0 (Baseline):** Standard answer-key feedback with correct option indicators.
- **Arm A_1 (Fixed Verified):** Complete, verified step-by-step worked solutions.
- **Arm A_2 (Mastery Adaptive):** Standard BKT mastery-based question sequencing.
- **Arm A_3 (Epistemically Governed Misconception Adaptive):** First-error localization → epistemic evidence reliability scoring → uncertainty-triggered diagnostic probing → minimal hint ladder → productive retry → targeted near-transfer.

12.1 Core Ablations and Target Metrics

The study will explicitly ablate:

1. **Direct LLM State Writes vs. Epistemic Evidence Governance:** Measuring calibration error, Brier score, and misconception macro- F_1 .
2. **Always-Teach vs. Uncertainty-Triggered Probing:** Measuring probe efficiency, time-to-stable-diagnosis, and student dropout rate.
3. **Resource-Aware Routing vs. Always-LLM:** Measuring latency, cloud inference cost, and local battery impact.

Primary learning outcomes will be assessed through pre-test/post-test normalized learning gains, delayed retention tests (at 4 and 8 weeks), error recurrence rates, and time-to-repair metrics.

13 Conclusion

Our prior reasoning engine successfully demonstrated that neuro-symbolic separation eliminates arithmetic hallucination, but also proved that automated problem solvers cannot serve as effective tutors. GonitSathi-BCS redirects these technological assets into an **Epistemically Governed Intelligent Tutoring System**. By treating student observations as verified evidence claims rather than immediate model writes, actively probing diagnostic uncertainty, enforcing pedagogical guardrails against answer leakage, and executing via resource-aware mobile routing, GonitSathi establishes a testable research trajectory centered on epistemic evidence governance, diagnostic uncertainty reduction, and verified pedagogical action.

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